SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 3 – Technical Presentation(5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 3: Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
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Section / Batch:	B/2024	**Date:**	21/10/26

Code of Conduct

I _____ *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Technical Presentation

Q1. A shopping mall uses CCTV cameras that produce clear images during the day but dark and noisy images at night, making it difficult to identify people and objects.
Prepare a technical presentation demonstrating how filtering, noise removal, contrast enhancement, and sharpening techniques can be applied to improve the captured images. Present the sequence of image-processing operations required before object detection.

Q2. A university is developing an automated document-scanning system using a camera, where captured documents may contain shadows, noise, uneven illumination, tilted pages, and blurred text.
Prepare a technical presentation explaining how filtering, thresholding, contrast enhancement, edge detection, and segmentation can be applied to improve the quality of the document images. Demonstrate the role of each technique in the processing pipeline.

Q3. A fruit-processing company wants to automatically identify damaged apples from images captured on a conveyor belt, where shadows, reflections, different backgrounds, and brightness variations are present.
Prepare a technical presentation comparing global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation for this application. Demonstrate how a suitable technique or combination of techniques can improve fruit defect detection.

Q4. An autonomous vehicle uses a front-facing camera to detect lane markings under different weather and lighting conditions, where shadows, damaged markings, vehicles, and rain may affect the captured images.
Prepare a technical presentation demonstrating how grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and region-of-interest techniques can be applied for lane detection. Explain the challenges associated with each processing stage.

Q5. A municipal corporation wants to classify waste such as plastic, paper, and metal using a camera installed above a conveyor belt. The captured images may contain overlapping objects, shadows, reflections, noise, and different object orientations.
Prepare a technical presentation demonstrating a complete image-processing pipeline using preprocessing, enhancement, segmentation, feature extraction, and classification. Explain how the proposed pipeline can be applied to handle real-world variations and improve waste classification.

Q1

Night-Time CCTV Image Enhancement

Scenario: Mall CCTV footage is clear by day but dark and noisy at night. The aim is to improve visibility while preserving details needed for reliable person and object detection.

Objectives

Reduce sensor noise without erasing important edges.
Improve brightness and local contrast in dark regions.
Sharpen useful details before object detection.

Recommended processing sequence

Input frame → Denoising → Contrast / brightness → Sharpening → Detection

Techniques and purpose

Noise removal: apply a median filter for salt-and-pepper noise or a Gaussian/bilateral filter for general noise. Avoid excessive smoothing because faces and clothing edges may disappear.
Contrast enhancement: use CLAHE (Contrast-Limited Adaptive Histogram Equalization) on the luminance channel to reveal details in dark areas without over-amplifying the whole image.
Brightness correction: use gamma correction with $\gamma < 1$ to brighten underexposed regions. Check highlights to avoid washing out lamps and reflective surfaces.
Sharpening: apply an unsharp mask or a mild Laplacian-based method after denoising. Sharpening before noise removal can make noise more visible.
Object detection: pass the enhanced frame to a detector, then compare detection confidence and missed/false detections against the original frame.

Evaluation and cautions

Inspect before/after crops and measure image quality using contrast, noise level, and detection precision/recall.
Enhancement cannot recover information that was never captured. Use adequate exposure, infrared illumination, and camera settings where possible.

Q2

Document Image Cleanup and Text Extraction

Scenario: Camera-scanned pages may have shadows, uneven lighting, noise, skew, and blurred text. The goal is a clean, aligned, high-contrast page suitable for OCR (Optical Character Recognition).

Processing pipeline

Capture → Denoise → Illumination / contrast → Deskew → Threshold → Edges & segmentation → OCR

Role of each technique

Filtering: use a median filter for isolated specks or a small Gaussian filter for sensor noise. Keep the kernel small so thin strokes remain intact.
Illumination and contrast: estimate the background illumination and normalize it, or use CLAHE to improve faint text. This reduces the effect of shadows and bright spots.
Deskewing: detect the dominant text-line angle using Hough lines or projection profiles, then rotate the page to make text lines horizontal.
Thresholding: global Otsu thresholding works when lighting is fairly uniform; adaptive/local thresholding is usually better when shadows or illumination gradients vary across the page.
Edge detection: Sobel or Canny can highlight page boundaries, text strokes, and table lines. Edge maps support alignment and layout analysis but are not a replacement for the document image.
Segmentation: separate the page into text blocks, lines, words, tables, and figures. Connected components and morphology can help, but touching characters may need more advanced separation.

Expected result and checks

Output: straight, high-contrast document with fewer background artifacts and clearer text regions.
Evaluate with OCR character/word error rate and check that punctuation, thin fonts, and table borders have not been removed.
Use perspective correction when the camera views the page at an angle; crop the page before OCR.

Q3

Apple Defect Detection: Segmentation Comparison

Scenario: Apples move on a conveyor with shadows, reflections, varied backgrounds, and changing brightness. Segmentation should isolate each apple and expose damaged or discoloured regions.

Compare the main methods

Method	Strength	Limitation / suitable use
Global thresholding	Fast and simple when fruit and background intensities are clearly separated.	Fails with shadows, reflections, and uneven lighting; use only with controlled lighting.
Adaptive thresholding	Computes local thresholds, handling gradual brightness changes better.	Can create noisy masks on textured skin or strong reflections; tune window size and offset.
Edge-based segmentation	Canny/Sobel edges can outline fruit and visible cracks or bruises.	Edges may break or be confused by stems, glare, and background texture.
Region-based segmentation	Groups similar colour/texture pixels; seeded region growing or clustering can isolate defects.	Sensitive to seed choice and colour variation; needs cleanup and parameter tuning.

Recommended combined pipeline

Colour correction → Denoise → Apple mask → Defect regions → Feature check → Classify

Control lighting and use a matte, contrasting conveyor background. Normalize brightness and apply a mild colour-space conversion (for example, HSV or Lab).

First isolate the whole apple using background colour/shape, adaptive thresholding, or a trained segmentation model. Remove small objects and fill holes with morphological operations.

Within the apple mask, detect candidate defects using local colour/texture differences, adaptive thresholds, and edge cues. Region-based grouping can merge connected damaged pixels.

Extract defect area ratio, colour difference, texture, and shape; classify as acceptable or damaged using rules or a trained model.

Validation

Test on images with varied fruit colours, lighting, orientations, and defect types. Measure segmentation IoU and defect precision/recall.

Q4

Camera-Based Lane Marking Detection

Scenario: An autonomous vehicle must detect lane markings despite shadows, worn paint, nearby vehicles, rain, and changing daylight. Classical image processing can provide lane candidates for a later tracking stage.

Processing pipeline

RGB frame → Grayscale → Filter → Contrast → Edges → Threshold → ROI → Lane fit

Stages and challenges

Grayscale conversion: simplifies processing by reducing three colour channels to intensity. Limitation: yellow/white lane colours can become similar to road brightness; colour cues may be retained in parallel.

Filtering: use a small Gaussian filter to suppress sensor noise and rain speckles. Strong smoothing can erase thin or worn lane markings.

Contrast enhancement: use CLAHE or local contrast normalization to make faint paint visible. Excessive enhancement can amplify road texture, shadows, and water reflections.

Edge detection: Canny detects strong intensity boundaries; tune thresholds to balance missed worn lines against false edges from vehicles, cracks, and barriers.

Thresholding: select bright pixels or combine intensity with colour masks. A fixed threshold is fragile across tunnels, sunlight, and wet roads; adaptive thresholds may be more robust.

Region of interest (ROI): keep the road area, often a trapezoid in the lower image, to reduce irrelevant edges. Incorrect ROI geometry can miss curves, hills, or nearby lane splits.

Lane fitting and tracking: fit candidate points with a line/curve model and track across frames. Reject implausible lane width or sudden position changes.

Robustness measures

Use temporal tracking, vehicle/road masking, and confidence scores. Rain and glare may require polarization, better optics, or a learned perception model.

Evaluate on day/night, wet/dry, shadowed, curved, and partially occluded lanes; report detection rate and lateral-position error.

Q5

Conveyor-Belt Waste Classification

Scenario: A camera must classify plastic, paper, and metal despite overlap, shadows, reflections, noise, and changing object orientation. A practical system combines image processing with a trained classifier.

Complete pipeline

Image capture → Preprocess → Enhance → Segment objects → Extract features → Classify → Sort / verify

Pipeline design

Capture and preprocessing: use fixed camera geometry, consistent exposure, and a plain conveyor background. Apply colour correction and a median or bilateral filter to reduce noise while preserving boundaries.

Enhancement: normalize brightness and use local contrast enhancement when needed. Avoid aggressive enhancement that changes the material's natural colour or makes metal glare look like a separate object.

Segmentation: threshold the background, then use morphology to remove small specks and close gaps. Connected-component analysis separates isolated items; overlapping objects may require watershed segmentation or an instance-segmentation model.

Feature extraction: calculate colour histograms, shape descriptors, area, perimeter, aspect ratio, texture, and edge density. Metal often reflects strongly, paper may have fibrous texture, and plastic appearance varies widely—no single feature is sufficient.

Classification: start with a baseline such as SVM or Random Forest using extracted features. For larger labelled datasets, a CNN can learn robust material features directly from image crops.

Decision and actuation: assign class and confidence; send low-confidence or heavily overlapped items to a second view or manual review. Synchronize the predicted class with conveyor position before activating a sorting mechanism.

Handling real-world variation

Train and test using varied backgrounds, lighting, rotations, crumpled paper, transparent plastic, dirty materials, and partially hidden objects. Use augmentation for rotation and brightness changes.

Track performance by class using confusion matrix, precision, recall, and throughput. Inspect errors separately for overlap, glare, and transparent materials.

Safety and reliability: include an emergency stop, guarded actuators, and a reject path for uncertain classifications.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____